

Qatar-4b

R-2683

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-4_210829 · created 2026-08-31T00:19:22+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459455.7815 +/- 0.003 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.179 +/- 0.053
Transit depth (Rp/Rs)^2	3.22 +/- 1.89 [%]
Orbital Inclination (inc)	89.09 +/- 2.17
Airmass coefficient 1 (a1)	26.74 +/- 25.06
Airmass coefficient 2 (a2)	-3.0 +/- 1.58
Scatter in the residuals of the lightcurve fit is	170.51 %
Best Comparison Star	None
Optimal Aperture	4.89
Optimal Annulus	6.93
Transit Duration (day)	0.0445 +/- 0.0043

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.179 $\pm$ 0.053 vs 0.138 (deviation 0.47 σ)
Duration fit vs published	0.0445 d vs 0.0892 d (deviation 6.38 σ)
Mid-time O-C	6.9 min
Depth SNR	2.1
Residual scatter	170.51 %

Light curve

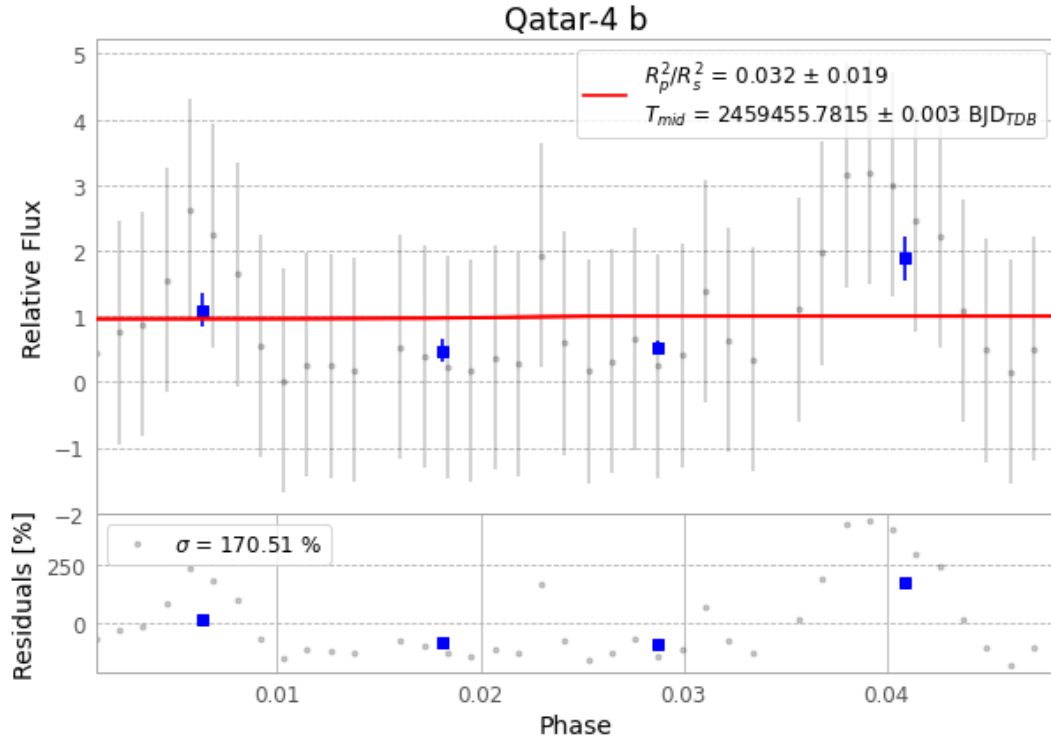


Figure 1 — FinalLightCurve_Qatar-4 b_2021-08-28.png (SHA-256 39a5e5aa33b474ae...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [358, 201], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1ee3f355964f1739...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

42 FITS frames (28 MB), Qatar-4210829064212.FITS → Qatar-4210829084512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-4-210829.log (339825973019...), FinalLightCurve_Qatar-4 b_2021-08-28.png (39a5e5aa33b4...), FinalLightCurve_Qatar-4 b_2021-08-28.csv (eae78dec1862...)

Compute resources

Wall time 169.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-31 00:19Z.